

Jesse Taube

🏠 Embedded Software Engineer

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🐙 GitHub Profile

🌐 LinkedIn Profile

🌐 My website

TECHNICAL SKILLS AND INTERESTS

Languages: C, C++, Bash, Verilog, Python, Rust

Tools & Technologies: Git, KVM, Podman, Linux Kernel, U-Boot, Buildroot, Debian

Protocols & Libraries: Networking, C&C++ STL, I²C, SPI, RS232

Hardware: PCB design, Soldering, Experienced with reading schematics, using oscilloscopes

Areas of Interest: Embedded Linux, IOT and IOT security

EXPERIENCE

• Rivos Inc

May 2024 - Sep 2024, May 2025 - Aug 2025

Kernel Developer

Remote

- RISC-V architecture improvements in Linux Kernel
- Python and Bash based test fixtures
- Custom Debugging hardware in QEMU and Riscv-isa-sim
- Implementing SBI extensions in Rust

• D3 Engineering

May 2023 - Aug 2023

Intern Kernel developer on the Nvidia Jetson Team

Rochester, New York

- Linux Kernel driver development in C for GMSL and FPD-link cameras
- Writing device trees and device tree overlays
- Reading schematics and datasheets
- Working with oscilloscopes and multi-meters
- Debugging hardware and software issues, such as MIPI CSI data loss

• Part-time Freelancer

Jun 2018 - Present

Hardware and Software developer

Hybrid

- Writing Kernel drivers for custom hardware
- Development of custom test boards
- UL-certified SOM-based custom motor control board with HDMI
- Programming tutor for University and high school students

PERSONAL PROJECTS

• Linux Ports

Porting of IMXRT and RP2350(Raspberry Pi Pico 2) to the Linux Kernel

- Writing Device Trees and Kernel drivers in C
- Debugging over JTAG with GDB
- Using agile development with co-developers

• KISC-V

A KISS principle RISC-V32I CPU, written in SystemVerilog

- Written in SystemVerilog and debugged with GTKWave
- Used Vivado and Quartus to run on Xilinx and Altera FPGAs
- Microcode generation in Python
- Custom trap handler in C and Assembly to emulate CSR instructions
- Can run No-MMU Linux

• IOT Lights

Reverse engineered wireless lights to integrate with Google Home

- Locally hosted Node based Google Home cloud app
- Custom hardware to control lights and communicate with cloud app
- Reversed engineering of Google Home's "actions on Google" API
- Reversed engineering of wireless protocol

EDUCATION

• University of Massachusetts Lowell

2022-26

Bachelor of Computer Engineering, Minor in Mathematics